11) Stochastic Gradient Descent

```
import numpy as np
X = np.array([1,2,3,4,5])
Y = np.array([2,4,6,8,10])
w = 0
b = 0
lr = 0.01
epochs = 50
for epoch in range(epochs):
    for i in range(len(X)):
        Y_pred = w * X[i] + b
        dw = -2 * X[i] * (Y[i] - Y_pred)
        db = -2 * (Y[i] - Y_pred)
        w = w - lr * dw
        b = b - lr * db
print("Weight:", w)
print("Bias:", b)
```

OUTPUT

Weight: 2.9452301495256 9
Bias: 0.23197802190218808